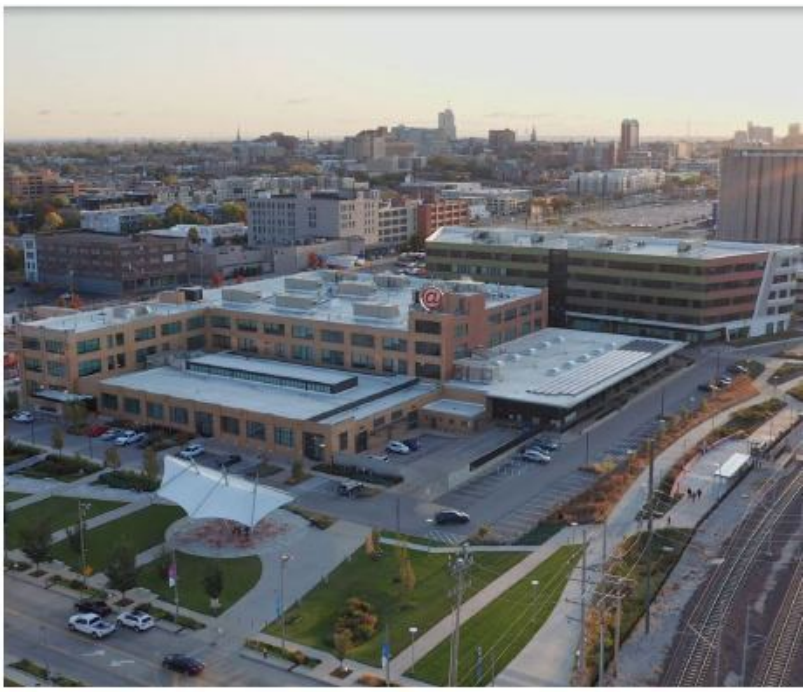
Green St. Louis

Four Projects Shaping The Future







Sculpting a Sustainable City Through Innovative Design

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Portfolio Overview:

This portfolio explores four key projects advancing sustainability in St. Louis:

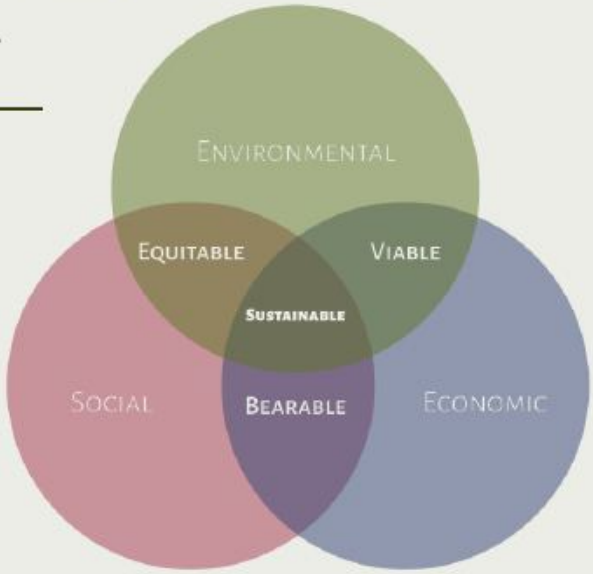
The +StL Greenway, the Jack C. Taylor Visitor Center, the Cortex Innovation District, and the St. Louis Zoo WildCare Park

In this portfolio, I will examine:

1. The history and design of each project
2. How their success aligns with UN Sustainable Development Goals
3. Measurable outcomes and opportunities for improvement.

Criteria for selection:

The four projects were selected for their alignment with the three pillars of sustainability—environmental, social, and economic. Each project demonstrates a diverse approach to stewardship, equity, and viability. Together, they represent an all-purpose effort to advance St. Louis as a model for sustainable development.



What are Sustainable Development Goals?

The United Nations' Sustainable Development Goals (SDGs) are a set of targets designed to address issues like poverty, inequality, and climate changes. With a large emphasis on the three pillars mentioned above, these 17 goals aim to create more sustainable and inclusive communities across the globe.



Introduction and History

Overview: A 50-mile urban renovation and landscape masterplan connection parks and waterways across the St. Louis Metropolitan area.

Commission: Led by Great Rivers Greenway (GRG) to enhance mobility, public health, and equitable access to green spaces

Origins:
Conceptualized as part of the River Ring plan, which is a large-scale framework to unite all parks, rivers, and communities in the region

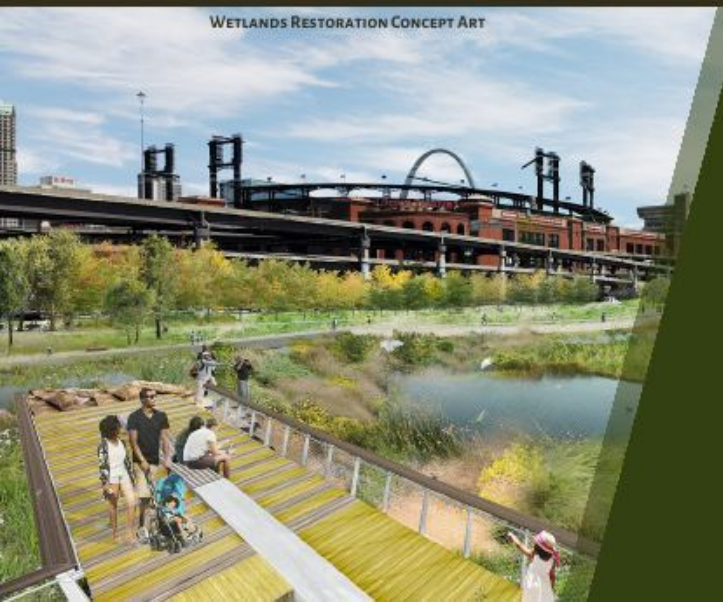
Development Timeline:
Initial planning began in the early 2000s, gaining traction amongst the public in 2019 when Stoss Landscape Urbanism won a competitive design bid to shape the project's future.







+STL Greenway



Design Team:

TLS Landscape Architecture	*Lead Designer*
Derek Hoeferlin Design	*Lead Designer*
Stoss Landscape Urbanism	*Design Competition Winner*
OBJECT TERRITORIES	*Collaborator*

Approach:

The project prioritizes sustainability by focusing on the three E's—environment, equity, and economy. They plan to create accessible greenways, restore ecosystems, and strengthen community through thoughtful, innovative design.



Aerial View of Proposed Greenway





Concept Art of Project

Success Highlights

Ecological Loops

Stormwater Management

Green Infrastructure

Equitable Extensions

Increased Accessibility

Cultural Preservation

Economic Assets

Revitalized Housing

Tourism and Events

Metrics of Success

+ Increase in native biodiversity

+ Expanded pathways for pedestrians and cyclists

+ Economic growth and revitalized spaces

↓ Reduced stormwater runoff and flooding

Room for Improvement

Collect and report measurable data for management successes

Place more emphasis on connections in underserved neighborhoods

Develop clear benchmarks to track economic and cultural growth

The +StL Greenway represents a sustainable vision for the future of St. Louis, uniting communities, enhancing accessibility, and promoting the incorporation of new green spaces across the city. By revitalizing abandoned areas and creating connections between our parks, neighborhoods, and landmarks, the project is a model for inclusive and environmentally concious urban spaces.

Jack C. Taylor Visitor Center

At the Missouri Botanical Garden



Designed to reimagine the visitor experience at one of the oldest botanical gardens in the nation, this LEED Gold-certified center embodies a sustainable approach to both architecture and landscaping. The project replaced the outdated Ridgeway Center in 2023, and now sees more than one million annual visitors.



Covered Outdoor Seating Area



Facade of the Emerson Conservatory

How it was Designed

The design team, led by Ayers Saint Gross and Michael Vergason Landscape Architects, prioritized accessibility, sustainability, and visual cohesion in the project. The building has a transparent southern facade to mimic greenhouses, while Missouri-native limestone and granite were used for durability and longevity. Six acres of woodland and grassland gardens hosting over 46,000 plants from 300 species surround the center.



Design Team:

Ayers Saint Gross

Michael Vergason Landscape Architects

Arbolope Studio

Jeffery Bruce & Co.

What Makes it Successful

The center successfully integrates key features of sustainability like energy-efficient design, immersive and biodiverse gardens that harbor endangered species, and utilization of natural light. The team incorporated sloping pathways and barrier-free amenities to ensure accessibility, and the Emerson Conservatory provides both sustainable landscaping and education on global ecology.

Room for Improvement

Expanded Educational Programming

Increased Community Outreach and Involvement

Aim for LEED Platinum Certification or Net-Zero Energy Certification

On-Site Renewable Energy Sources

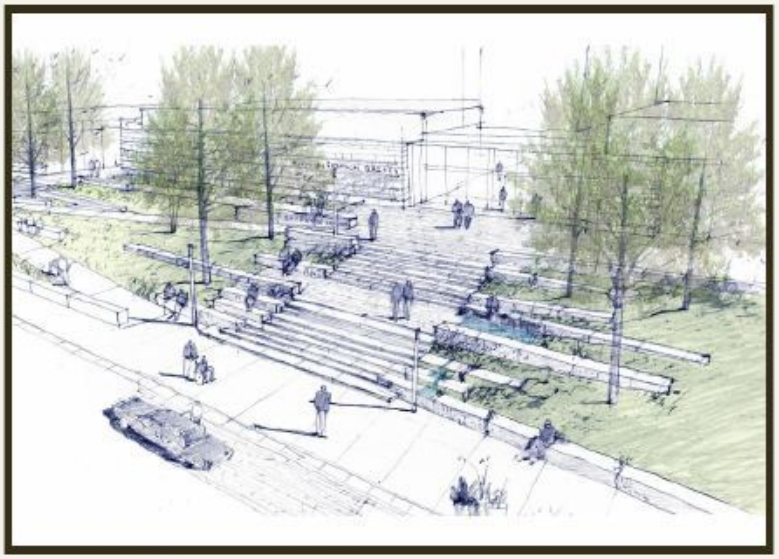
Success Metrics

Visitor Numbers and Satisfaction

Plant Collection Diversity

LEED Gold Certification Requirements Met





The Jack C. Taylor Visitor Center is a beautifully designed gateway to the Missouri Botanical Garden that has seamlessly blended sustainability, accessibility, and education. Its innovative, LEED Gold Certified design and immersive, conservation-forward gardens set a new standard for green architecture and visitor experience.

Cortex Innovation District



Aerial View of the Cortex Innovation District

The Cortex Innovation District, launched in 2002, transformed 200 acres of neglected industrial land into a hub for entrepreneurship and innovation. The project is backed by institutions, like Washington University and BJC Healthcare, and has become a large driver in the St. Louis economy, attracting a grand total of $885M in investments as of 2023. The district's design prioritizes community and efficient infrastructure, while blending with sustainability and aesthetic. The mixed-use district is now home to over 400 companies, including Microsoft, and has generated more than 6,000 jobs.



@4240 Duncan: The HOK-designed, LEED Platinum Workplace

Design and Construction

HOK, a top-ranked global architecture firm, and Clayco, a design-build leader, spearheaded the district's development. HOK's designs blend adaptive reuse with new construction, emphasizing on collaboration-friendly spaces like the Cortex Commons and LEED-certified buildings like 4220 Duncan and @4240 Duncan.

Clayco largely managed construction, focusing on sustainable practices and delivery. Key projects from them include the CIC (Cambridge Innovation Center), @4240 Duncan, and the BJC Institute of Health. Infrastructure upgrades were also implemented, like the MetroLink Cortex Station, to enhance accessibility, walkability, and offer more green spaces.

Development Partners

HOK	Lead Architecture Firm
Clayco	Design-Build Firm
Arcturis	Design and Planning
SWT Design	Landscape Architecture

Collaborators

City of St. Louis

Washington University in St. Louis

BJC Healthcare

University of Missouri–St. Louis (UMSL)

Saint Louis University (SLU)

Missouri Botanical Garden

Successes and Impact Of Cortex



Cortex Commons: Outdoor recreation area for all members

Environmental Impact

- Supports 52 plant species and 23 pollinators
- Exceeds stormwater storage needs by 21%
- The District is home to 5 LEED-certified buildings

Social Impact

- Home to 400+ startups and organizations
- On-site programs have trained over 1,200 entrepreneurs
- Outdoor commons are regularly used by 55% of the workers

Economic Impact

- Generated $2.1 billion in economic output for St. Louis
- Created over 5,000 jobs since its inception
- Has contributed around $1B to local development



4220 Duncan | 2018



4340 Duncan | 2019

Potential Improvements

- Attract more sustainability-focused ventures
- Increase the usage of renewable energy systems on site
- Expand the number of LEED-certified buildings
- Develop affordable housing options nearby for the workers

Key Takeaways

The Cortex Innovation District stands as a premier example of urban revitalization and innovation-driven development. By blending sustainable infrastructure, economic opportunity, and social equity, it has transformed a neglected part of St. Louis into a bustling hub for innovation. Landmark achievements include the multiple LEED-certified buildings, the vast green space, and the updated MetroLink station.

Cortex extends further than business as well, with biodiversity and stormwater systems being largely successful. Public collaboration spaces like Cortex encourage community involvement and lead to forward-thinking development. The project is just one of the first steps toward creating a more sustainable city.



St. Louis Zoo WildCare Park







Concept Art of Visitor Experiences

About the Park

The St. Louis Zoo's WildCare Park is a significant expansion of the current wildlife conservation and education provided by the Zoo. Located in northern St. Louis County, the new 425 acre park plans to connect visitors with wildlife through immersive experiences, while promoting biodiversity and conservation efforts.

How it Started

The STL Zoo purchased the property in 2018 to progress its conservation efforts. By 2021, the zoo had conceptual plans and artwork, showing the guest experiences that will be offered. Construction has been steady and efficient, with the park expected to be open to the public in 2027.



Contributors

Saint Louis Zoo

The Kent Family

City of St. Louis

Achievements and Future Directions

Areas of Success

Expanded land area for wildlife capacity

Birth of baby animals happening on-site

An estimated 400k+ guests annually

With the St. Louis Zoo, a combined economic impact of $2.5 billion over the next decade

Over 250 animals expected to live in the park



Critically Endangered Addax Calf, Born On Site | September 2024

Potential for Improvement

Incorporate educational programs like the Zoo has

Promote large-scale conservation efforts

Improve public transportation options to the park

Integrate a large variety of visitor engagement opportunities



Concept Art for Open Giraffe Feeding Exhibit

The St. Louis Zoo WildCare Park is a magnificent step forward in both education and conservation, allowing the current zoo to improve on their animal habitats while increasing their preservation abilities. With the large 425 acre facility, it will make a significant impact in modern ecological conservation. The park is estimated to be successful for all three pillars of sustainability, and will greatly help in connecting people with nature and global conservation efforts.

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